SAMPLE

Great Observatories: Gaia

Mapping a Billion Stars

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Introduction

Ask a stranger to name the most productive scientific instrument ever built, and the answers will run to particle colliders, space telescopes with golden mirrors, and laboratories the size of cities. The correct answer is stranger than any of them: a two-ton spinning cylinder, launched to a quiet spot far out in the direction of the night, that has spent more than a decade doing one old-fashioned thing, measuring the positions of stars, better than anyone in history, for nearly two billion of them.

Gaia is the European Space Agency's astrometry mission, and astrometry, the science of the positions and motions of celestial bodies, is the oldest science in the world. Every ancient surveyor of the sky, from Hipparchus counting the stars for his catalogue to Tycho Brahe aiming his brass at the planets, was an astrometrist. The science's great problem was always the same: the stars are far away, and far-away things barely move, and the tiny shifts that reveal their distances and journeys are buried in a noise of instruments and atmosphere. For two thousand years, astrometry crept forward one decimal at a time. Gaia, named in a roundabout way for the Greek goddess of Earth, moved it forward three decimals in one generation, from the thousandth of a degree to the billionth.

What that leap bought is the subject of this book. With distances to a billion stars, the Milky Way stopped being a picture and became a machine: its spiral arms, its streams of shredded satellite galaxies, its cycles of stellar birth and death, all suddenly measurable, because nearly everything about a star becomes interpretable once you know where it is and how it is moving. With motions, the galaxy acquired a history that can be run backward: Gaia found the fossil of a collision with another galaxy buried in the orbits of millions of stars, a discovery that changed what the Milky Way is. With repeated measurements, the mission watches the sky change: starquakes, passing asteroids, wobbles of stars around unseen companions including, remarkably, the black holes that no other method could find. And with its archive of everything, published openly for the whole world, Gaia became not just an instrument but the substrate on which a generation of astronomy now runs, a foundation cited, in one way or another, by more scientific papers each year than any other project in existence.

The name is a collision of meanings, and this book sorts them out honestly. There is the Greek goddess, the primordial mother of earth and sky, whose myth properly belongs to other shelves. There is the Gaia of Lovelock's hypothesis, the proposal that the biosphere behaves like a self-regulating organism, an idea in earth science that also belongs elsewhere. And there is Gaia the spacecraft, whose name began as a faint acronym, Global Astrometric Interferometer for Astrophysics, and survived its own redesign to become, conveniently, an invocation of the ancient goddess of the sky's own subject. This volume is about the third Gaia: the mission, its ancestors, its machinery, its catalogues, its discoveries, and the revolution in our knowledge of the galaxy that its numbers have driven.

The chapters follow a simple plan. The first four tell the story of what Gaia is and how it works: the mission's origins in its heroic predecessor Hipparcos, the plain explanation of astrometry and parallax, and the strange, spinning, one-dimensional telescope that performs the measurements. The middle chapters cover the working life: the scanning choreography, the ground stations and the thousand-person consortium that turns star transits into a catalogue, the successive data releases, and the mission's place in the era of big astronomy. The later chapters assess the harvest: the discoveries, the engineering struggles, the open mysteries, and what comes after, including the missions and instruments being built to chase what Gaia found. Technical ideas are explained as they arrive, in plain words, and no mathematics is required. What is required is the willingness to be amazed by a simple thing done very, very well.

One warning belongs at the threshold, in fairness to the reader who met the other Gaias first. Search the word and the goddess comes first, the hypothesis second, and the spacecraft third, which is exactly backwards by productivity. The mission has, at this book's writing, generated more than ten thousand scientific papers, mapped more stars than there are humans alive, and fixed the sky's coordinate system for a century; the myth generated the myth, and the hypothesis generated a debate. Each belongs to its own book, and the goddess, gracious parent of both the name and the family census, can surely spare the spacecraft its turn. The reader who wants the myth or the hypothesis will find other shelves; the reader who wants to know how humanity mapped the galaxy will find it here, in the story of the third Gaia, the one with a billion-pixel eye.

A final orienting fact sets the scale for everything that follows. The mission's founders compare its catalogue to the effect of the first photographic surveys of the sky a century before, or the effect of the first star catalogues of

antiquity: each of those instruments changed astronomy by making the sky's inventory bigger and better, and each took generations to exploit. The difference is speed, since the modern catalogue is open, digital, and global: the exploitation of the billion-star map began the day it was published, by everyone at once, and it is proceeding, as the reader will see, at a pace the earlier surveys never knew. This book is a snapshot of that exploitation, taken at full speed, with the history of the machine and the reasoning of the science explained for anyone who wants to understand, not just use, the greatest map of the galaxy ever made.

Chapter 1: The Discovery of Gaia

No space mission is born alone, and Gaia's story begins with its ancestor, Hipparcos, the satellite that proved the idea could fly.

The concept of measuring star positions from space was old when spaceflight was young. The French astronomer Pierre Lacroute proposed a space astrometry mission in the 1960s, understanding a truth the rocket age made concrete: the atmosphere, which makes stars twinkle, makes them wander too, shuffling a star's apparent position by a hundredth of a second of arc and more, forever blurring the fine measurements. A telescope above the atmosphere, observing stars continuously, could do better in a year than ground observatories had managed in centuries. The European Space Agency, young and bold, adopted the idea as its first dedicated science mission, named, with a classical flourish, for Hipparchus, the ancient Greek who founded astrometry and, in the second century before our era, compiled the sky's first surviving star catalogue.

Hipparcos launched in 1989, and nearly died at once. A motor failure stranded it in the wrong orbit, an elongated ellipse instead of the intended circle, exposed to radiation belts its electronics had never been designed for. The rescue, one of the great improvisations of the space age, saved the mission: engineers reprogrammed the satellite to observe through the radiation, ground stations around the planet caught its data as it swept past, and the satellite, against every prediction, produced its full sky survey by 1993. The Hipparcos Catalogue, published in 1997, contained the positions, parallaxes, and proper motions of over a hundred thousand stars, precise to about a thousandth of a second of arc, along with a companion survey of millions more at rougher precision. Astronomers had waited their whole careers for it. In the years that followed, Hipparcos was cited in what seemed like every paper on stars, and the question at every conference became the same: when do we fly the better one?

The better one was being designed even before Hipparcos failed. In the early 1990s the scientists who had run the first mission, Michael Perryman above all, working with colleagues across Europe including the British astronomer Gerald Gilmore and the Swedish astrometry master Lennart Lindegren, drafted the concept of a successor with two commitments: measure fainter stars, vastly more of them, and measure them ten to a hundred times better. The trick, as chapter four explains, was not a bigger mirror, which astrometry does not need, but a bigger field of view, longer observing time, and colder engineering discipline over angles and stability. The concept was proposed to the Agency in the mid-1990s as the Global Astrometric Interferometer for Astrophysics, a design built around optical interference in the spirit of the era's radio interferometers; the design matured, the interferometer gave way to a pair of ordinary, superbly engineered telescopes, and the acronym, conveniently already spelling the name of the Greek goddess, stayed. Gaia was formally approved as a cornerstone of the Agency's science program in 2000, its baseline commitments now set at measuring a billion stars to a precision measured in millionths of a second of arc.

The build took a decade. Industrial teams in Europe, under the Agency's management, constructed a spacecraft like no other: a three-mirror telescope design duplicated into two identical telescopes pointed 106.5 degrees apart, sweeping a focal plane of a hundred and six large CCD chips, nearly a billion pixels, the largest camera ever flown in space. A ten-and-a-half-meter sunshield, folded like an umbrella around the satellite for launch, would keep the whole structure in permanent shade and thermal silence. Micro-thrusters burning cold gas, not rockets, would steer, because even the puff of a conventional thruster's plume would shake the measurements. The spacecraft, weighing about two tons, was one of the most precisely engineered objects ever made, and its instruments were calibrated on the ground to tolerances measured in atoms.

Gaia launched on a Soyuz rocket from Europe's spaceport at Kourou, on the coast of South America, on the nineteenth of December 2013, and sailed for a month toward its station: the second Lagrange point, a gravitational balance point a million and a half kilometers in the Sunward direction, where the spacecraft could orbit the Sun in formation with our planet, its back to the daylight, its telescopes in permanent cool shadow. Commissioning in the first half of 2014 revealed the usual surprises, a faint stray light in the payload, some contamination of the mirrors, a thruster or two behaving oddly, but also confirmed the fundamental design: the instrument worked, and its first

star transits measured angles the way the simulations had promised. Routine surveying began in July 2014. The mission was cleared to run its nominal five years; it would, in the end, run for more than eleven.

The approval story deserves one more beat, because it explains the mission's scale. When the Agency committed to Gaia in 2000, the proposal's headline numbers, a billion stars, microarcsecond precision, seemed to many in the community like the usual inflation of the grant-writing genre; the mission's scientists themselves, in candid retrospectives, admit that the billion was more vision than calculation, a stake driven into the future's ground. The engineering then had to grow toward the stake. The camera's pixel count, the downlink's data rate, the processing consortium's size, all multiplied several-fold during the design decade, as the teams discovered, stage by stage, what a billion stars actually cost; and the mission that flew in 2013 was, by the standards of the original sketches, a different animal, larger, colder, and more disciplined. The pattern is a lesson in how big science actually happens: the dream sets the target, and the engineering, a decade of it, negotiates the price.

The launch night itself deserves its paragraph, because it fixed the mission's date with the heavens. December in Kourou is the rainy season's edge, and the Soyuz, imported to the spaceport for exactly such medium-class payloads, rose on the nineteenth of 2013 with the spacecraft folded inside its fairing, the sunshield wrapped around the telescope like a winter coat. The ascent was nominal, the departure toward the distant station precise, and within days the mission's engineers were watching the first star transits cross the focal plane, the first measurement of the ten-year harvest. Veteran team members, who had spent two decades on the project, some of them from Hipparcos's own rescue generation, record the moment in their memoirs with a restraint that reads like awe: the spacecraft, after a month's cruise, opened its survey, and the numbers began.